

Zihao Jin

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📍 Shenzhen, Guangdong, China | 🎯 Research: Multimodal LLMs, Computer Vision, Robotics

Education

Sun Yat-sen University

School of Artificial Intelligence

Guangdong, China

Sep. 2023 – Expected Jun. 2027

- GPA: 86/100. Coursework: Computer Vision (97), Machine Learning (96), Human-Computer Interaction (94), Numerical Computing (96), Digital Signal Processing (94), Game Theory (90).
- Honors: SYSU Outstanding Student Scholarship; Provincial Mathematical Modeling Award; MCM Honorable Mention; 3rd Place, National Collegiate Dragon Boat Championship; Excellence Award, “Win at SYSU” Competition.

Research & Project Experience

Nereus: A Large-Scale Underwater Dataset for Fine-Grained Attribute

Nov. 2025 – May 2026

Understanding and Grounded Counting

Research Project, NeurIPS 2026 Submission

- Built and cleaned an underwater multimodal dataset with 88K images, 2.68M QA pairs, 136K objects, and 65K point annotations; helped design a 136-dimensional object-level attribute taxonomy.
- Constructed region-aware counting, camouflage-aware counting, and density-hotspot reasoning tasks from point annotations; evaluated Qwen, InternVL, LLaVA, and related models for benchmark analysis.

EvoSkill-Agent: Self-Evolving Physical Skill Acquisition for Embodied

May 2026 – Present

Agents

Research Project, Embodied AI and Humanoid Robot Learning

- Proposed an experience-driven embodied skill learning framework that transforms robot trajectories, visual observations, and execution feedback into reusable physical skills, addressing the limitation of current VLA models that learn actions but lack lifelong skill accumulation.
- Designed an embodied skill memory and evolution pipeline integrating GR00T-WholeBodyControl with editable diffusion policy modules, enabling skill discovery, composition, failure diagnosis, and iterative refinement for Unitree G1 manipulation tasks.

Ego2Robot: Egocentric Data Collection and Imitation Learning

May 2026 – Present

Pipeline for Unitree G1

Research Project, Humanoid Robot Data Pipeline

- Built a Unitree G1 teleoperation pipeline with RealSense D435i, PICO/NoMachine, and GR00T-WholeBodyControl, enabling synchronized image streaming and manipulation data collection.
- Established a real-robot imitation learning workflow for cup-grasping tasks, covering data cleaning, LeRobot/GR00T format alignment, fine-tuning preparation, and evaluation.

Skills

Languages & Tools: Python, C++, MATLAB, Linux, Git, Gradio

ML / Vision: PyTorch, OpenCV, multimodal LLMs/VLMs, Diffusion Policy/EDP, object detection, pose estimation

Robotics / Data: Isaac Lab, GR00T, LeRobot, RealSense, ZMQ, data collection, benchmark construction, model evaluation

Additional Information

- Languages: Mandarin Chinese (native); English: CET-4 / CET-6.
- Research strengths: multimodal dataset construction, embodied robot data collection, model evaluation, and cross-disciplinary application exploration.